AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC | Retrofits

OUR PROJECTS ACROSS INDIA

Water Treatment | Wastewater Treatment | ZLD | Industrial Solutions

Pune, Maharashtra



SITE IN PROGRESS

STP – 250 KLD (MBBR Technology)
Civil + Electro Mechanical

Ludhiana, Punjab



SITE COMPLETED

RO Plant – 15,000 LPH
MGF + ACF + Softener + UF + RO + UV

Gurugram, Haryana



SITE IN PROGRESS

ETP – 100 KLD (MBBR)
MSRL Tanks + Electro Mechanical

Baddi, Himachal Pradesh



SITE COMPLETED

Flouride Removal Plant – 20 m^3/hr
MSEP Vessels with Activated Alumina Media

Solan, Himachal Pradesh



SITE COMPLETED

UF + RO Plant – 8,000 LPH
For Industrial Process Water

Ludhiana, Punjab



SITE COMPLETED

ZLD Plant after ETP – 30 KLD
UF + RO + MEE + ATFD

Pune, Maharashtra



SITE IN PROGRESS

ETP – 75 KLD (Physico-Chemical + MBBR)
Installation and Piping Work

Gurugram, Haryana



SITE COMPLETED

RO Plant – 10,000 LPH
MGF + ACF + Softener + RO + UV

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Office & Factory :
Sr. No. 165/2/9, Alankapuram Road,
Opposite Yashobhumi Society, Adarsha Nagar,
Charholi Budruk, Pimpri-Chinchwad,
Maharashtra – 412105, India.

+91- 9622554445 / +91 - 9284544150 / +91 - 9226197281
ceo@aquenfluid.com / sales@aquenfluid.com
www.aquenfluid.com

WATER | WASTEWATER | ZLD | AMC

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Towards Zero Liquid Discharge For a Cleaner Greener Tomorrow

ZERO LIQUID DISCHARGE (ZLD) PLANT

Complete Solution for Industrial Wastewater Treatment & Water Recovery



Maximum Water Recovery | Environmental Compliance | Cost Optimization | Customized Design | Reliable Operation | End-to-End Support

WHAT IS ZERO LIQUID DISCHARGE (ZLD) ?

Zero Liquid Discharge (ZLD) is an advanced wastewater treatment system designed to achieve maximum possible water recovery, with no liquid effluent discharged outside the plant. The treated water is recycled and reused in the process, and the concentrated brine is converted to solid salts through evaporation and crystallization.

KEY BENEFITS

- No liquid discharge to environment
- High water recovery (90–95%)
- Compliance with environmental norms
- Reduced freshwater consumption
- Recovery of reusable water & valuable salts
- Suitable for high TDS and complex industrial effluents
- Automated and reliable operation
- Customized design as per industry and water quality

TYPICAL PROCESS FLOW DIAGRAM (ZLD AFTER ETP)



MAJOR UNITS IN ZLD PLANT

PRE-TREATMENT



- Pressure Sand Filter (PSF)
- Activated Carbon Filter (ACF)
- Chemical Dosing System
- FRP / MSEP / MSRL vessels

ULTRAFILTRATION (UF)



- Removal of suspended solids
- Protects RO membranes
- High flux PVDF membranes
- Automatic backwash system

REVERSE OSMOSIS (RO)



- High recovery RO (70–85%)
- SS-304/316 skids and piping
- Energy efficient design
- CIP system for long life

EVAPORATION SYSTEM (MEE)



- Multi Effect Falling Film Evaporator
- High energy efficiency
- Suitable for high TDS effluent
- SS-316 construction

ATFD / CRYSTALLIZER



- Agitated Thin Film Dryer (ATFD)
- Salt crystallization to dry solid
- Minimum effluent discharge
- PLC based automatic operation

APPLICATIONS



Pharmaceutical Industry | Chemical Industry | Textile Industry | Electroplating Industry | Food & Beverage Industry | Dairy Industry | Automotive Industry | Common Effluent Treatment Plants (CETP)

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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TECHNICAL SPECIFICATIONS & STANDARD MODELS

Model	Capacity (m³/hr)	Treated Water Recovery (%)	RO Reject to MEE (m³/hr)	Steam Consumption (kg/hr)	Power (kW)	Solid Salt (kg/hr)*
ZLD-5	5	90–95	0.25–0.50	150	60	40
ZLD-10	10	90–95	0.50–1.00	300	100	80
ZLD-20	20	90–95	1.00–2.00	600	180	160
ZLD-30	30	90–95	1.50–3.00	900	250	240
ZLD-50	50	90–95	2.50–5.00	1500	400	400
ZLD-100	100	90–95	5.00–10.00	3000	750	800

* Solid salt quantity is indicative and depends on influent TDS.

INLET & OUTLET WATER QUALITY (TYPICAL)

Parameter	ETP Treated Water (Feed to ZLD)	Recovered Water (RO Permeate)
pH	6.5 – 8.5	6.5 – 8.5
TSS	< 10 mg/L	< 1 mg/L
COD	< 100 mg/L	< 10 mg/L
BOD	< 30 mg/L	< 5 mg/L
TDS	2,000 – 20,000 mg/L	< 100 mg/L
Conductivity	3,000 – 30,000 µS/cm	< 200 µS/cm

CONSTRUCTION OPTIONS

FRP SYSTEM



- Corrosion resistant FRP vessels
- Suitable for small to medium capacity
- Compact and lightweight
- Easy installation
- Cost effective

MSEP SYSTEM (Mild Steel Epoxy Painted)



- Mild steel with epoxy internal coating
- Suitable for medium to large capacity
- High strength and long life
- Custom design as per requirement
- Cost effective for industrial application

MSRL SYSTEM (Mild Steel Rubber Lined)



- Mild steel with rubber lining
- Suitable for high TDS and aggressive effluents
- High corrosion resistance
- Long service life
- Custom fabrication

TYPICAL LAYOUT OF ZLD PLANT



PROJECT PHOTOGRAPHS (ONGOING & COMPLETED)



COMPLETED PROJECT
Pharmaceutical Industry – Gujarat
ZLD Plant – 30 m³/hr



ONGOING PROJECT
Textile Industry – Rajasthan
ZLD Plant – 20 m³/hr



COMPLETED PROJECT
Electroplating Industry – Punjab
ZLD Plant – 50 m³/hr



COMPLETED PROJECT
Chemical Industry – Bihar
ZLD Plant – 25 m³/hr

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ceo@aquenfluid.com / sales@aquenfluid.com

www.aquenfluid.com

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